VDI 2221 and VDI 2225: System Integrity and Governance in Engineering Design

Section 1: The Foundations of Systematic Design in the VDI Framework

1.1 Defining Systematic Engineering Design: Principles and Objectives

The German guideline VDI 2221, formally titled "Design of technical products and systems - Model of product design," establishes a structured, methodical blueprint for modern product development. This standard provides central guidelines for interdisciplinary application across various industrial practices by defining core objectives, specific activities, and expected work results.[1] Its foundational premise is that successful product development relies on an applied systematic approach, flexibly adapted to the task at hand.[2]

The scope of VDI 2221 is intentionally broad, applying to all classifications of technical products and systems, as well as the accompanying interdisciplinary design and development processes.[1] The design methodology formalized in VDI 2221 is deeply rooted in systems engineering, problem-solving techniques, and cybernetics. This scientific orientation was pursued specifically to provide a more rigorous, quantifiable foundation for design practice.[3]

Critically, the standard's target audience extends far beyond the traditional mechanical engineer. VDI 2221 explicitly addresses process managers, project development managers, product managers, and various process stakeholders, alongside product designers and academics.[1] This broad recipient list confirms that the standard's function is not merely methodological instruction—the

how to design—but rather a comprehensive framework for **process governance**. By standardizing the design methodology, VDI 2221 ensures procedural consistency, facilitates robust auditability, and supports efficient knowledge transfer across large, complex, and

interdisciplinary organizations, which is paramount for managing scale in modern product development.

1.2 Historical Trajectory: From Individual Genius to Corporate System

The guidelines published by the Association of German Engineers (VDI) represent a deliberate and coherent evolution of design methodology over half a century. The lineage of VDI 2221 can be traced back to early guides, such as Fritz Kesselring's 1954 "Guideline for invention," which was based heavily on his personal practical experiences and focused on externalizing conditions for successful individual invention.[4]

Subsequent generations of guidelines, including those developed by Hansen and the precursor VDI 2222, progressively moved away from focusing on the individual's creative process toward a focus on systematic thought and the system idea.[3] This shift broadened the standard's aim from merely supporting efficient inventive processes to supporting the complete design process within the integrated environment of a company.[3] The intention changed from providing personal support for individual thinking and action to establishing a general procedure that addresses organizational structure and content.[3]

Accompanying this change in focus was a fundamental change in the level of instruction. Early guidelines offered firsthand, immediately actionable advice, such as suggestions to study patent literature or write down ideas. VDI 2221, however, operates at a higher, more abstract level, providing instructions that must be adapted to the specific current situation of the designer.[3] For example, instead of instructing a designer to study literature, VDI 2221 mandates the establishment of a formal function structure or structure modules.[3] This critical movement toward abstract, process-oriented terminology and instructions proactively positioned the German systematic design methodology for integration with modern technology. Abstract concepts like "establishing a function structure" are more readily formalized into algorithms or process steps, facilitating the essential integration of "research results and computer support in design".[3] This abstraction was therefore a necessary precondition for effective Computer-Aided Engineering (CAE) support and the transition to digital engineering workflows.

1.3 Structure and Interrelation of VDI 2221 (Part 1 & 2) and VDI 2225

VDI 2221 is generally understood as the overarching methodological framework, describing

the principal procedure in its Part 1, the "Model of product design".[1] VDI 2221 Part 2 provides configuration details for individual product design processes, elaborating on objectives, requirements, and interfaces required for practical application.[5]

The overall hierarchy of the VDI guidelines dictates that VDI 2221 defines the core systematic *Vorgehensweise* (methodology), while later, specific guidelines delve into the procedures within particular phases.[7] VDI 2225, though often discussed alongside 2221, acts as a dedicated companion standard for one of the most high-risk steps in the development cycle: objective concept evaluation and selection. The structure ensures that the general, holistic guidance provided in VDI 2221 is supported by detailed, phase-specific methodology standards.

Section 2: Task Clarification and Requirements Specification (Phase I)

2.1 The Foundational Phase: Clarify and Define the Task

Phase I, the clarification and definition of the task, is the foundational stage upon which all subsequent design activities rest. It begins with the initial product planning, which may be triggered by external factors such such as market demand and environmental shifts, or by internal corporate impulses generated by marketing or product management departments.[8]

To effectively clarify the task and obtain the necessary functions, designers must conduct extensive data acquisition. This involves going beyond conversations with the immediate client to incorporate feedback from customer requests, analyzing complaints and technician reports, reviewing sales reports concerning lost orders, and examining competitor solutions and patent literature.[8]

The systematic methodology stresses that the process is not linear. VDI 2221 suggests continuous adaptation and iteration of the requirements list throughout the entire design process.[3] This inherent iterative imperative requires designers to loop both forward and backward between preceding and subsequent stages as new knowledge is acquired.[3] A critical action at this stage is the

abstraction of the task. This technique is specifically recommended to help designers

overcome fixation on initial solutions or prejudices, ensuring the conceptual design starts with an unbiased, function-oriented perspective.[3]

2.2 Formal Requirements Management: Lastenheft versus Pflichtenheft

A defining feature of the German systematic approach, formalized by VDI 2221, is the mandatory and rigorous separation of the requirements documentation into two distinct formal documents: the *Lastenheft* (Demand Specification) and the *Pflichtenheft* (Functional Specification).[9] This distinction is critical for maintaining process clarity and managing stakeholder expectations.

The *Lastenheft* defines the client's requirements—*what* is required of the final product—and typically originates from marketing or product management. Conversely, the *Pflichtenheft* defines the engineering team's proposed solution—*how* the requirements will be fulfilled.[9] The

Pflichtenheft serves as a detailed technical charter for the development effort. It must elaborate on the functional and technical architecture of the intended solution, including the functional structure, the effective structure (*Wirkstruktur*), essential system elements, and, critically, the explicit definition of all interfaces (*Schnittstellen*) connecting these elements.[6]

The necessity of formally separating these documents elevates the initial stage into a **formal contractual negotiation** between the market/client side and the R&D side. The R&D team is prohibited from proceeding into solution generation until the external demands (Lastenheft) have been translated into a set of internal, verifiable technical constraints (Pflichtenheft). This procedural discipline, combined with the mandated continuous iteration of the specification [3], provides the necessary organizational structure to incorporate new technical knowledge while minimizing scope creep and managing external expectations.

The relationship between these key documents is formalized as follows:

Table 1: Comparative Roles and Content of Lastenheft and Pflichtenheft (VDI 2221)

Document (German/English)	**Primary Focus (The Question)**	**Typical Ownership/Author**	**Core Content Elements**	**VDI Phase Link**
Lastenheft (Demand	What needs to	Client, Marketing, or	Objectives, Constraints,	Input to Phase I (Task

Specification)	be achieved?	Product Management	Boundary Conditions, Desired Results, Quality expectations.	Clarification)
Pflichtenheft (Functional Specification)	How will the requirements be met?	Development/ Design Engineering Team	Function Structure, Physical Principles, Defined Interfaces (Schnittstellen) , Verification Criteria, Technical Architecture.	Output of Phase I (Formalized Requirements)

Section 3: Conceptual Design and Solution Synthesis (Phase II)

3.1 Function-Oriented Analysis and Abstraction

Phase II of the VDI 2221 model, conceptual design, transforms the clarified task into a set of potential solution principles. The core activities involve determining the system's function structure and initiating the search for ways to realize those functions.[3]

This process relies heavily on function decomposition. The overarching required function is broken down into constituent subfunctions, thereby creating a precise function hierarchy.[10] This hierarchical breakdown allows the designer to isolate the specific function elements required to perform necessary unit transformations.[10] For complex systems, such as advanced manufacturing coating processes, the methodology suggests supplementing standard system analysis with a function-oriented analysis. This approach ensures that the links between

desired object properties and the necessary technological manufacturing parameters are rigorously established.[11]

3.2 Generating and Combining Solution Principles (The Creative Step)

VDI 2221 prescribes the systematic assignment of specific design methods to each phase.[3] The conceptual phase is designated as the stage for maximum creative divergence. Creative methods are deployed here with the explicit goal of generating a wide, comprehensive range of different solution variants.[12]

Key creative methods employed in this phase include traditional brainstorming and Method 635, alongside the indispensable **Morphological Analysis**.[12] Morphological analysis allows for the combinatorial generation of solutions by listing possible means for fulfilling each subfunction and assembling them into composite variants.[10] The total solution is synthesized by systematically combining these function elements.[10]

This seemingly chaotic process of divergent thinking is systematically managed by the VDI framework through strict functional anchoring. While creativity is encouraged, all generated solutions must be traceable back to the function structure established in Phase I.[3] Advanced computational tools, such as the Multilayer Functional Structure (MFF), extend this concept. MFF, based on a mathematical model, generates functional structures that simulate the iterative nature of the conceptual phase, ensuring that new concept variants are systematically derived from functional requirements.[11] This systematic management of divergence ensures that creativity remains constrained by the defined requirements, preventing the development of solutions that address the wrong problem.

Section 4: Objective Evaluation and Selection using VDI 2225

4.1 The Role of VDI 2225 in Decision Gate Management

The systematic approach dictates that decisions regarding the selection of the "best"

concept variant must be objective and quantifiable to justify the ensuing resource commitment. VDI 2225 provides the formal methodology to achieve this objectivity, primarily through the technique known as Use-Value Analysis, or *Nutzwertanalyse*.[13]

VDI 2225 formalizes the process of determining a product concept's value, assessing its usefulness and strength relative to the defined project objectives.[13] The methodology is founded on the principle that the value of a solution is not absolute; rather, it is measured against pre-defined, rigorously weighted parameters derived directly from the

Pflichtenheft.[13]

This formal evaluation process transforms VDI 2225 into a powerful **formal negotiation and risk-prioritization tool**. By mandating that management, marketing, and engineering agree on the relative *importance* (weighting) of criteria—such as performance, cost, and manufacturability—*before* concept evaluation begins, the standard guarantees that all stakeholders share a quantifiable definition of the "optimal" solution. This ensures that the convergence to a single concept is auditable, justifiable, and based on objective metrics rather than subjective opinion.

4.2 Application of the Value Scale and Criteria

The quantitative comparison of different concepts requires standardized tools. VDI 2225 provides a specific Value Scale (*Werte-Skala*) to ensure that the scoring of each concept against the weighted criteria is consistent and comparable.[14] This process converts qualitative expert assessments into quantifiable, objective data, forming the final basis for the decision to proceed with a selected concept.[13]

A crucial component of modern VDI application is the integration of long-term strategic factors. The evaluation technique mandates that assessments must consider economic and ecological effects early in the design phase.[15] This proactive requirement, often aligned with related standards like VDI 4800 (Resource Efficiency), ensures that sustainability and lifecycle cost metrics are built into the design choice from the outset, rather than being addressed as expensive modifications later in the development process.[15]

The fundamental structure and requirements of the *Nutzwertanalyse* are summarized below:

Table 2: Principles of Concept Evaluation (Nutzwertanalyse) based on VDI 2225

Evaluation Element	Description/Purpose	VDI 2225 Requirement	Impact on Design Process
Weighted Parameters	Criteria derived from the *Pflichtenheft* (e.g., cost, reliability, efficiency, manufacturability).	Parameters must be pre-defined and assigned relative weighting factors based on project objectives.[13]	Forces early prioritization and quantification of customer and technical needs.
Value Scale	A standardized numerical scale used to score how well each concept fulfills a weighted parameter.	Ensures scoring is consistent and quantifiable across different concept variants.[14]	Converts subjective expert opinion into objective, comparative data.
Balanced Value Profile	The summation of weighted scores for all parameters, yielding a total utility value.	Serves as the quantitative basis for the objective decision to proceed (convergence).[13]	Provides clear justification for design choices to management and auditors.

Section 5: Embodiment and Detail Design (Phases III and IV)

5.1 Phase III: Embodiment Design (Conceptual Layout)

Following the objective selection of the optimal solution principle using VDI 2225, Phase III focuses on translating that principle into a concrete, physical architecture—the embodiment design. The primary instruction in this phase is to subdivide the selected principle into **realizable modules**.[3]

This involves rigorous attention to the physical realization of the solution. The intended

solution is broken down into main groups and essential system elements (partial systems), accompanied by the precise, unambiguous definition of all connecting links, or interfaces (*Schnittstellen*).[6] The explicit requirement to define modules and interfaces is a direct application of systems engineering principles to mechanical design. This systematic decomposition facilitates subsequent simultaneous engineering efforts, allowing parallel development workstreams for different modules, and fundamentally establishes the architectural basis necessary for effective

configuration management later in the product lifecycle.

This phase results in the development of preliminary layouts and, subsequently, the definitive overall layout of the key modules and the complete product.[3]

5.2 Phase IV: Detail Design and Final Documentation

Detail design is the final step in the VDI 2221 process, focused on generating all necessary information required for production, assembly, operation, and maintenance. This phase includes specialized tasks, such as the detailed calculation of structural components, using the VDI 2221 methodology to analyze loads and determine specifications, as demonstrated in practical applications.[16]

The culmination of Phase IV is the preparation of production and operating instructions, leading to the creation of the definitive Product Documents.[3] While the VDI 2221 flowchart illustrates a sequence of phases and steps (determine functions, search for principles, develop layouts, prepare instructions) [3], the methodology explicitly requires final validation and verification steps that close the iterative loop and confirm that the final product meets the specifications established in the

Pflichtenheft.[16] The comprehensive documentation generated at this stage forms the essential input for downstream Product Lifecycle Management (PLM) systems.

Section 6: Theoretical Context and Comparison with Pahl & Beitz

6.1 VDI 2221 as the Standardized Pahl & Beitz Model

The systematic approach formalized by VDI 2221 is inseparable from the influential work of Pahl and Beitz, whose 'Systematic Approach' is internationally recognized as the standard textbook description of methodical design.[2] The Pahl and Beitz model is generally understood as a prescriptive model, derived from meticulous observations of successful professional design practice.[18]

VDI 2221 effectively acts as the standardized codification of this prescriptive framework, providing the official German industrial guidance for the methodology described in the standard textbook.[7] The VDI framework and the Pahl & Beitz model share the same core structure: breaking the design process down into four principal phases (task clarification, conceptual design, embodiment design, and detail design).[2] This shared, systematic approach serves a critical pedagogical function, providing mechanical engineering students with a foundational structure for their early design education.[18]

6.2 Prescriptive Capacity versus Predictive Reality

A major area of academic investigation involves testing the robustness of the systematic model, examining whether the prescriptive VDI/Pahl & Beitz model can actually *predict* the behavior of designers. Studies often use protocol analysis, coding the cognitive actions of students using schemas like Function–Behaviour–Structure (FBS) to compare real-world design behavior against the idealized model.[18]

Empirical findings suggest that the Systematic Approach predicts *some* but notably **not all** of student design issue behavior.[18] This observation does not constitute a failure of the VDI model. Rather, it emphasizes a crucial philosophical distinction: design, as a human activity, inherently involves non-linear cognitive processes, often characterized by intuitive heuristics and spontaneous breakthroughs that deviate from a strict, sequential flow.

Therefore, the systematic VDI model is best interpreted not as a perfect descriptive map of human cognitive processes, but as the **rational structure for the documentation and control of design**. It dictates the required output structure—the systematic evidence, decision gates, and formal documentation—regardless of the designer's internal, heuristic journey. This focus ensures that the design process, even if cognitively non-linear, yields verifiable, systematic results suitable for organizational use, regulatory compliance, and archival purposes.

The close relationship and minor operational differences between the VDI standard and the widely accepted textbook model are evident in the following comparative assessment:

Table 3: Methodological Comparison: VDI 2221 vs. Pahl & Beitz Systematic Approach

Design Characteristic	**VDI 2221 (Model of Product Design)**	**Pahl & Beitz (Systematic Approach)**	**Key Distinction/Convergence**
Core Structure	Four primary phases, emphasizing organizational integration and continuous feedback loops.[3]	Four macro-phases, often taught as sequential, but inherently iterative.[2]	High Convergence: VDI formalizes the Pahl & Beitz structure into a certifiable standard.[7]
Requirements Definition	Explicit separation and mandatory documentation of **Lastenheft** (what) and **Pflichtenheft** (how).[6]	Strong emphasis on the formulation and adaptation of a comprehensive requirements list.	VDI 2221 enforces the specific document hierarchy required by German industry and legal standards.
Focus Evolution	Developed to support general company processes, systems engineering, and interdisciplinary application.[3]	Developed from observations of successful professional mechanical design practice.[2]	VDI 2221 broadened the scope from individual design thinking to holistic, organized corporate procedures.[3]

Section 7: Strategic Implementation and Future Directions

7.1 Strategic Integration in Contemporary Engineering

The robust, structured nature of the VDI framework allows it to manage highly specialized, interdisciplinary efforts. The methodology has proven applicable across diverse fields, demonstrated in case studies ranging from the structural design of pump test equipment frames [16] to the complex, iterative development of advanced biomedical systems.[16]

The systematic approach requires the inherent utilization of established methods crucial for high-level product development, such as the mandatory deployment of morphological analysis for exhaustive concept generation during Phase II.[12] Furthermore, the detailed documentation and definition of interfaces in Phase III establishes a sound architecture that supports simultaneous engineering and digital product development tools, including advanced virtual reality applications.[12]

A key strategic benefit of the VDI structure is its function as a tool for **risk mitigation and intellectual property (IP) asset structure**. The systematic generation of multiple design variants (Phase II) and their objective, quantifiable selection (VDI 2225) create a comprehensive, defensible audit trail. By formally documenting all considered alternatives, the objective evaluation criteria, and the precise reasons for the final choice, companies can scientifically defend their design decisions against legal challenges, liability claims, or competitive scrutiny. The formalized process ensures that the intellectual property generated is supported by rigorous procedural documentation.

7.2 Recommendations for Process Managers

The success of implementing VDI 2221 relies heavily on managerial interpretation and enforcement. The methodology should never be treated as a rigid, waterfall model. Process managers must recognize the **iterative imperative** inherent in the standard; the VDI flowchart, despite its sequential appearance, explicitly mandates iteration both toward and backwards between stages.[3] Therefore, project budgeting and scheduling must strategically allocate time and resources for these necessary feedback loops and for the continuous adaptation of the requirement specification.[3]

To optimize resource allocation and maintain objectivity, process managers should mandate the formal utilization of VDI 2225 at all major concept decision gates. Employing VDI 2225 is the primary defense against subjective bias, ensuring that the critical decision to commit

resources to Embodiment Design is based exclusively on quantifiable value assessment agreed upon by all internal stakeholders.[13]

7.3 Sustainability and Resource Efficiency

Modern engineering demands that design methodologies integrate holistic lifecycle assessment criteria. The VDI framework has proven adaptable to these requirements. By linking VDI 2221 to related standards, notably VDI 4800, the methodology supports the integration of criteria regarding resource efficiency, environmental impact, and economic effects.[15] This systematic inclusion of assessment criteria in the task clarification and conceptual phases ensures that resource efficiency is addressed proactively, leading to solutions that are inherently sustainable and economically optimized over their entire product lifecycle.[7]

Conclusions

VDI 2221 and VDI 2225 constitute the rigorous, formalized spine of systematic engineering design in advanced industrial contexts. VDI 2221 provides the essential architectural model, defining the four core phases of product development and shifting the focus from individual inventive action to organized, interdisciplinary process governance. This model's primary strength lies in its procedural discipline: mandating the formal separation of the *Lastenheft* and *Pflichtenheft* to convert external market demands into quantifiable technical constraints.

The companion guideline, VDI 2225, addresses the critical convergence phase by formalizing the objective selection of concept variants via *Nutzwertanalyse*. This evaluation method, requiring pre-weighted parameters, functions as a powerful mechanism for stakeholder alignment and risk prioritization, ensuring that decisions are auditable and grounded in scientific comparison.

The systematic approach, while sometimes challenged by empirical observations of heuristic design behavior, remains invaluable because it provides the required **rational documentation structure** for all organizational outputs. For R&D directors, the VDI framework is not merely a set of technical instructions, but a strategic asset that ensures process integrity, supports multi-disciplinary concurrency, validates design choices scientifically, and constructs a robust defensive structure for intellectual property throughout the entire product lifecycle. Consistent, disciplined application of VDI 2221 and VDI 2225 is essential for mitigating

development risk and optimizing the allocation of engineering resources in the creation of complex technical systems.

Works cited

1. VDI 2221 Blatt 1 - Design of technical products and systems - Model of product design, accessed September 30, 2025, https://www.vdi.de/en/home/vdi-standards/details/vdi-2221-blatt-1-design-of-technical-products-and-systems-model-of-product-design
2. Engineering Design - Apple Books, accessed September 30, 2025, https://books.apple.com/us/book/engineering-design/id483222008
3. THE DEVELOPMENT OF THE GUIDELINE VDI 2221 - THE CHANGE OF DIRECTION | The Design Society, accessed September 30, 2025, https://www.designsociety.org/download-publication/18983/THE+DEVELOPMENT+OF+THE+GUIDELINE+VDI+2221+-+THE+CHANGE+OF+DIRECTION
4. THE DEVELOPMENT OF THE GUIDELINE VDI 2221 - THE CHANGE OF DIRECTION | The Design Society, accessed September 30, 2025, https://www.designsociety.org/download-publication/18983/the_development_of_the_guideline_vdi_2221_-_the_change_of_direction
5. VDI 2221 Blatt 2 - Design of technical products and systems ... - Details, accessed September 30, 2025, https://www.vdi.de/en/home/vdi-standards/details/vdi-2221-blatt-2-design-of-technical-products-and-systems-configuration-of-individual-product-design-processes
6. Vdi-Richtlinien Verein Deutscher Ingenieure Entwicklung ... - Scribd, accessed September 30, 2025, https://de.scribd.com/document/642202238/Untitled
7. Konstruktionsmethodik, accessed September 30, 2025, https://www.ressource-deutschland.de/fileadmin/user_upload/3_Werkzeuge/a_Basis-Werkzeuge/i_Leitfaden_Ressourceneffizienz/Konstruktionsmethodik.pdf
8. P R O D U K T E N T W I C K L U, accessed September 30, 2025, https://www.hs-anhalt.de/fileadmin/Dateien/FB6/personen/voigt_st/Lehrunterlagen/08_PE/02-03_PE_Modellvorstellungen_und_Produktplanung.pdf
9. Pflichtenheft und Lastenheft - Weiterbildungen - VDI Wissensforum, accessed September 30, 2025, https://www.vdi-wissensforum.de/weiterbildungen/pflichtenheft-lastenheft/
10. Recommended design process (VDI 2221 [151]). - ResearchGate, accessed September 30, 2025, https://www.researchgate.net/figure/Recommended-design-process-VDI-2221-151_fig17_280940066
11. Design Methodology according to VDI 2221 [43,44 ,45] - ResearchGate, accessed September 30, 2025, https://www.researchgate.net/figure/Design-Methodology-according-to-VDI-2221-43-44-45_fig4_303838308
12. PRODUCT DEVELOPMENT METHODS IN VIRTUAL REALITY - Cambridge University Press, accessed September 30, 2025,

https://www.cambridge.org/core/services/aop-cambridge-core/content/view/CA7051CDE80D0E3303D5938A85FADCE5/S2732527X2100506Xa.pdf/product-development-methods-in-virtual-reality.pdf
13. Hannu Oja - Incremental Innovation Method for Technical Concept Development with Multi-disciplinary Products - Trepo, accessed September 30, 2025, https://trepo.tuni.fi/bitstream/handle/10024/114623/oja.pdf?sequence=2
14. Value scale for VDI guideline 2225 evaluation technique - ResearchGate, accessed September 30, 2025, https://www.researchgate.net/figure/Value-scale-for-VDI-guideline-2225-evaluation-technique_tbl1_363345867
15. Konstruktionsmethodik - VDI ZRE, accessed September 30, 2025, https://www.ressource-deutschland.de/leitfaden-re/methoden/konstruktionsmethodik/
16. The development of the guideline VDI 2221 - The change of direction - ResearchGate, accessed September 30, 2025, https://www.researchgate.net/publication/283412253_The_development_of_the_guideline_VDI_2221_-_The_change_of_direction
17. Pahl/Beitz Konstruktionslehre - 2021-01 - DIN Media, accessed September 30, 2025, https://www.dinmedia.de/de/publikation/pahl-beitz-konstruktionslehre/314700897
18. Can Pahl and Beitz' systematic approach be a predictive model of designing? - Cambridge University Press, accessed September 30, 2025, https://www.cambridge.org/core/services/aop-cambridge-core/content/view/8BFE1D692F764BCE7D0CEF539ED3C580/S2053470117000245a.pdf/can-pahl-and-beitz-systematic-approach-be-a-predictive-modelof-designing.pdf